



Figure 1: Rockstor installation



Figure 2: Rockstor installation—select date and timezone, and installation destination

some of the components you'll need:

1. Case (e.g., SilverStone DS380B)
2. Power supply unit (e.g., SS-350FE)
3. Motherboard and CPU combo (e.g., ASRock C2550D4I and the Intel Avoton C2550 or C2750 quad-core processor)
4. RAM (e.g., Kingston 8GB 1600MHz DDR3L DIMM)
5. Boot drive (a USB or a PCIe SATA)
6. HDDs (e.g., Western Digital hot pluggable — 8.89cm HDDs or 6.35cm HDDs)

Building NAS on pre-built servers: This approach is recommended for prosomers or SMBs. The advantage of this approach is its cost-effectiveness and flexibility in the choice of hardware (as long as it meets the minimum system requirements) or hypervisors.

You can use any of the following four options to build your storage system:

1. Old hardware such as an old laptop or an old server
2. Appliances such as Intel NUC, Asus Vivo PC, HP ProLiant, etc
3. Hypervisors such as KVM, Proxmox or VirtualBox
4. A combination of virtualisation and appliances

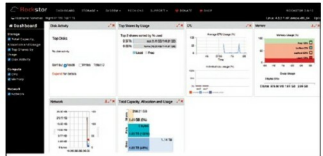


Figure 3: Rockstor dashboard

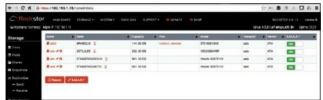


Figure 4: Create storage-available hard disks



A note on minimum system requirements: Each of the Linux NAS distros will have specific system requirements depending on the underlying file system.

- With Rockstor, you'll need the following:
- 64-bit Intel or AMD processor
- 2GB RAM or more (recommended)
- 8GB hard disk space for the OS
- One or more additional hard drives for data (recommended)
- Ethernet interface (with Internet access, for updates)
- A UPS system (if desired) that is supported by Network UPS Tools
- DVD drive and a blank DVD, or a USB port and minimum 1GB USB key (for the installation media)

Building a complete NAS solution from an open source

Linux NAS distro provider: This solution is recommended for SMBs or even prosomers who would like to save time and not mind paying the extra cost. The advantage of this solution is its plug-and-play nature. Rockstor Pro 8 DIY NAD Build is one such solution with BTRFS at its core.

This solution is superior to other Linux based solutions available in the market as users have access to advanced features at a fraction of the cost. Some of these features are:

- Simple and secure browser based management
- Supports different sized HDDs and has online capacity scaling. Adds more drives as and when you need them
- Bitrot protection, checksums, compression and other advanced file system (BTRFS) features
- Efficient Copy-on-Write (CoW) snapshots of shares (on demand or on schedule)
- File sharing and access from Linux, Mac, Windows and mobile devices (Android and iOS)